MTH 331(A) Dr. James Ashe
EXAM October 12, 2012 **Name:**

Justify your answers and show your work. All questions are of equal value.

1. Find an explicit formula for the general n^{th} term of the sequence, $\{a_n\}_{n=0}^{\infty} = \{1, 2, 4, 8, 16, \dots\}$.

2. Write the infinite series, $0.3 + 0.03 + 0.003 + \cdots$, using sigma notation (i.e. using Σ) .

3. Write the first 3 terms of the sequence of partial sums of the series $\displaystyle\sum_{k=1}^{\infty} \frac{3^k}{2}$.

Evaluate the geometric series or state that it diverges.

4. $\displaystyle\sum_{k=0}^{\infty} \frac{2^k}{7^k}$.

5. $\displaystyle\sum_{m=2}^{\infty} \left(\frac{1}{3}\right)^m$.

Find the limit of the following sequences or determine that the limit does not exist.

6. $\lim_{n\to\infty} \left\{ \dfrac{n^4 - 1}{n^4 + 10} \right\}$.

7. $\lim_{n\to\infty} \left\{ \dfrac{n^{0.01}}{\ln n} \right\}$.

Use the Divergence Test to determine whether the following series diverge, or state that the test is inconclusive (use your results from above).

8. $\displaystyle\sum_{k=1}^{\infty} \frac{k^4 - 1}{k^4 + 10}$.

9. $\displaystyle\sum_{k=1}^{\infty} \frac{k^{0.01}}{\ln k}$.

Using the Squeeze Theorem, find the limit of the following sequence or determine that the limit does not exist.

10. $\lim_{n\to\infty} \left\{ \dfrac{\sin^2 n}{2^n} \right\}$.

Use the Integral Test to determine whether the series converge, diverge, or state why the test cannot be used. Check that the conditions of the test are satisfied.

11. $\sum_{k=1}^{\infty} \dfrac{k}{e^k}$. For $f(x) = \dfrac{x}{e^x}$, $f'(x) = (1-x)\,e^{-1}$ and $\int f(x)\,dx = 2xe^{-1}$.

12. $\sum_{k=2}^{\infty} \dfrac{1}{(2x-3)^2}$. For $f(x) = \dfrac{1}{(2x-3)^2}$, $f'(x) = \dfrac{-4}{(2x-3)^2}$ and $\int f(x)\,dx = \dfrac{-1}{2(2x-3)}$.

Use the Ratio Test to determine whether the following series converges.

13. $\sum_{k=1}^{\infty} \dfrac{3}{k!}$.

Use the Root Test to determine whether the following series converges.

14. $\displaystyle\sum_{k=1}^{\infty}\left(\frac{k-1}{3k}\right)^{k}$.

Use the Comparison Test or Limit Comparison Test to determine whether the following series converge. HINT: $\sum \frac{1}{k^p}$ *converges for* $p > 1$ *and diverges for* $p \leq 1$.

15. $\displaystyle\sum_{k=1}^{\infty}\frac{1}{k^2+4}$.

BONUS: Use the formal definition of the limit of a sequence to prove that $\displaystyle\lim_{n\to\infty}\left(3+\frac{1}{n}\right) = 3$.